SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:	VIMALA K	**Reg. No.:**	192424276
Date:	6/8/26	**Duration:**	60 minutes

Code of Conduct

I VIMALA K 192424276 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student vimalak

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or technically

	major components, interfaces, and interactions accurately represented.		lacks sufficient detail.	incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	Mark
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25

SOFTWARE ENGINEERING — ARCHITECTURE DESIGN ANALYSIS

Intelligent Electric Vehicle Charging Network Management System For ARCHITECTURE DESIGN ANALYSIS

Student Name: VIMALA K

Register Number: 192424276

Department: B.Tech (AI&DS)

SLIDE 2

System Requirement Analysis

Problem Statement

Develop an Intelligent EV Charging Management System to monitor charging stations, predict waiting times, optimize charger allocation, manage charging sessions, and improve customer experience.

System Objectives

- Monitor charging stations in real time
- Predict waiting time
- Optimize charger allocation
- Generate energy reports
- Improve customer satisfaction

Functional Requirements

- User Registration and Login
- Search Charging Stations
- Book Charging Slot
- Start and Stop Charging Session
- Online Payment
- Notifications
- Energy Report Generation

Non-Functional Requirements

- Performance
- Security
- Reliability
- Availability
- Scalability
- Maintainability
- Usability

SLIDE 3

Software Architecture Selection

Selected Architecture

Hybrid Architecture (Client–Server + Layered Architecture)

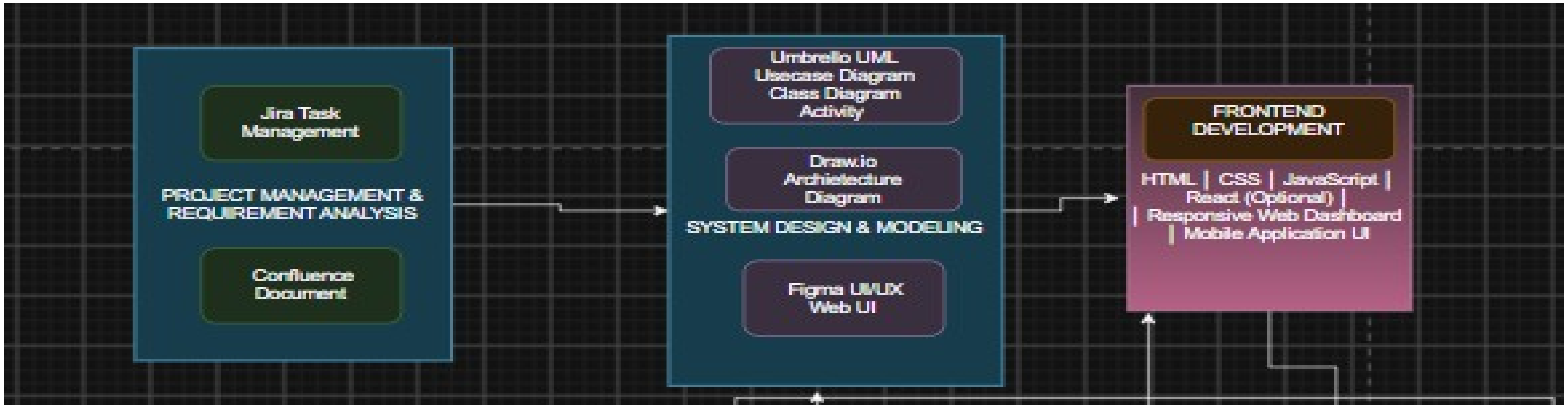
Justification

- Supports web and mobile applications
- Easy to maintain
- Secure communication through REST APIs
- Scalable for future charging stations
- Supports integration with external APIs
- Separates presentation, business logic, and database layers

SLIDE 4

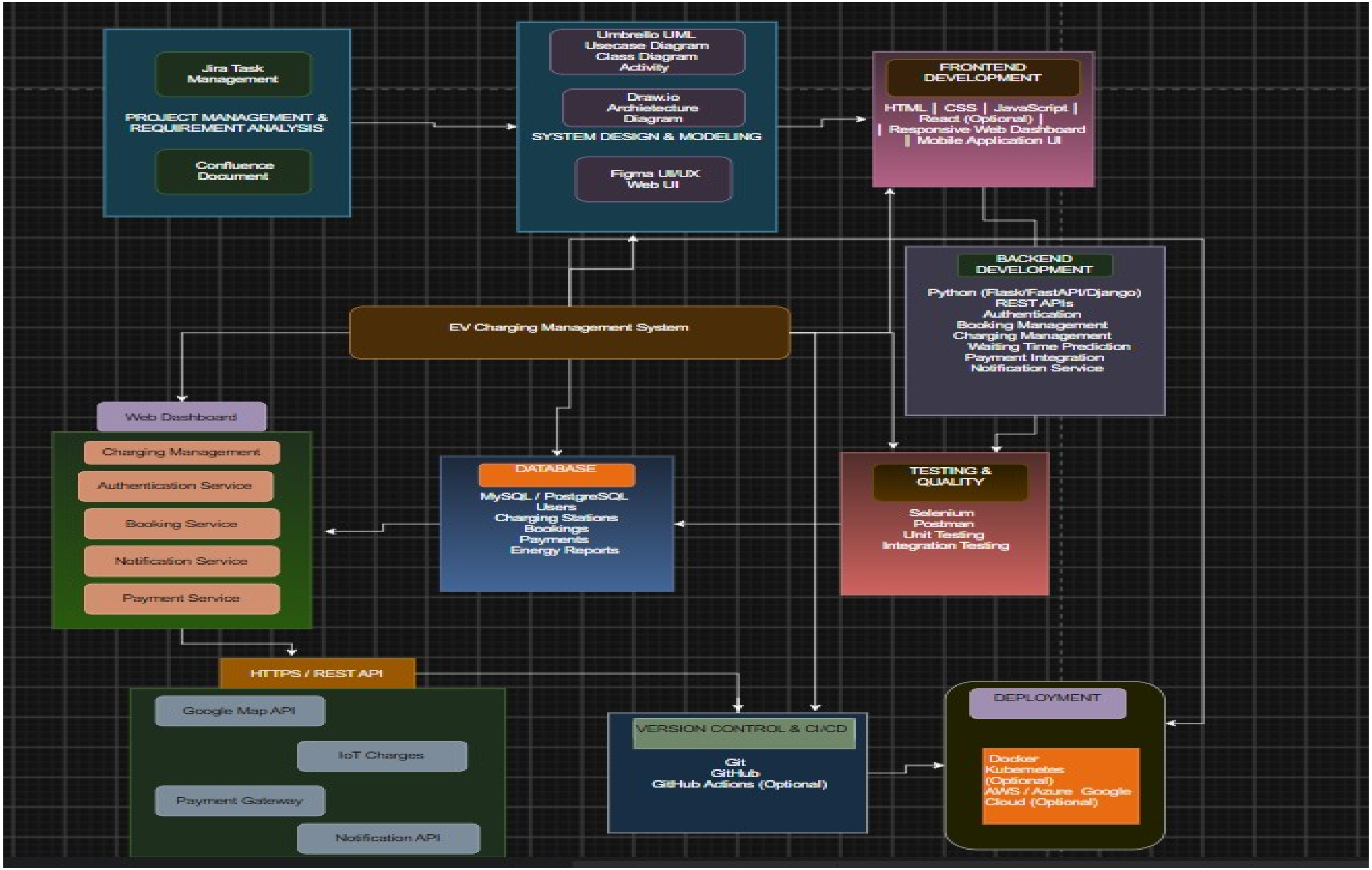
Software Architecture Diagram

Hybrid Architecture



Main Components

- Presentation Layer
- Application Layer
- Business Logic Layer
- Data Layer
- External APIs
- Database
- IoT Charging Stations



Jira Task Management
PROJECT MANAGEMENT & REQUIREMENT ANALYSIS
Confluence Document
Umbrello UML Usecase Diagram Class Diagram Activity
Draw.io Architecture Diagram
SYSTEM DESIGN & MODELING
Figma UI/UX Web UI
FRONTEND DEVELOPMENT
HTML | CSS | JavaScript | React (Optional) | | Responsive Web Dashboard | Mobile Application UI
BACKEND DEVELOPMENT
Python (Flask/FastAPI/Django) REST APIs Authentication Booking Management Charging Management Waiting Time Prediction Payment Integration Notification Service
EV Charging Management System
Web Dashboard
Charging Management
Authentication Service
Booking Service
Notification Service
Payment Service
DATABASE
MySQL / PostgreSQL Users Charging Stations Bookings Payments Energy Reports
TESTING & QUALITY
Selenium Postman Unit Testing Integration Testing
HTTPS / REST API
Google Map API
IoT Charges
Payment Gateway
Notification API
VERSION CONTROL & CI/CD
Git GitHub GitHub Actions (Optional)
DEPLOYMENT
Docker Kubernetes (Optional) AWS / Azure Google Cloud (Optional)

SLIDE 5

Components and Interactions

Presentation Layer

- Mobile Application
- Web Dashboard

Application Layer

- Authentication Service
- Booking Service
- Charging Management
- Payment Service
- Notification Service

Business Logic Layer

- Waiting Time Prediction
- Charger Allocation
- Energy Management
- Report Generation

Data Layer

- User Database
- Charging Station Database
- Booking Database
- Payment Database

Workflow

Users → Frontend → REST API → Backend → Database → External APIs → Response to User

SLIDE 6

Quality Attributes

Quality Attribute	Implementation
Scalability	Supports more charging stations and users
Security	Authentication and secure REST APIs
Performance	Fast response time and optimized queries
Reliability	Continuous monitoring and error handling
Maintainability	Modular architecture and easy updates
Availability	24x7 system access

Architectural Decisions

Why Hybrid Architecture?

- Supports both mobile and web applications
- Easy to expand in the future
- Modular design
- Better code maintenance
- Easy API integration
- Suitable for cloud deployment

Trade-offs

- Slightly higher development complexity
- More components to manage
- Requires API communication

Future Enhancements

- AI-based waiting time prediction
- Smart pricing
- Renewable energy integration
- Vehicle-to-Grid (V2G)
- Predictive maintenance

SLIDE 8

Conclusion

The Hybrid Architecture provides a scalable, secure, reliable, and maintainable solution for the Intelligent EV Charging Network Management System.

It Enables:

- Real-time monitoring
- Smart charger allocation
- Secure online payment
- Efficient energy management
- Better customer experience

Assignment Outcome

- ☑ System Requirements Analyzed
- ☑ Suitable Architecture Selected
- ☑ Architecture Diagram Prepared
- ☑ Components and Interactions Explained
- ☑ Quality Attributes Evaluated
- ☑ Architectural Decisions Justified

Thank You